

Disproof of Conjecture 00000009924

An edge insertion makes a Hodge eigenvalue jump

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The conjecture defines persistent Laplacians as Hodge Laplacians varying with a filtration parameter and asserts that their eigenvalues are Lipschitz in that parameter. For the ordinary unweighted Hodge Laplacians of a simplicial filtration, already the diagonal persistent case, a single edge insertion disproves this assertion.

Let the vertices be $0, 1$. For $t < 0$, let K_t contain the empty simplex and the two singleton vertices. For $t \geq 0$, also include the edge $\{0, 1\}$. These are downward-closed simplicial complexes and $K_s \subseteq K_t$ whenever $s \leq t$. Vertex degrees are at most one, and the only insertion is one edge.

Use the usual inner products in which oriented simplices are orthonormal. The edge chain space $C_1(K_t; \mathbb{R})$ is zero-dimensional before insertion and one-dimensional afterward; the vertex space is $\mathbb{R}^2$ throughout. Orient the edge from 0 to 1. Its boundary matrix is

$$B_t : C_1(K_t; \mathbb{R}) \longrightarrow \mathbb{R}^2, \quad B_t = \begin{cases} \text{the } 2 \times 0 \text{ matrix,} & t < 0, \\ \begin{pmatrix} -1 \\ 1 \end{pmatrix}, & t \geq 0. \end{cases}$$

The unaugmented degree-zero Hodge Laplacian is $L_t = B_t B_t^*$, since there is no lower boundary contribution. Consequently

$$L_t = \begin{cases} \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, & t < 0, \\ \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, & t \geq 0. \end{cases}$$

Before insertion, both eigenvalues are zero. After insertion, $(1, 1)$ is a zero eigenvector and $(1, -1)$ is a two eigenvector. These independent vectors span the entire vertex space, so the complete sorted eigenvalue list is $(0, 2)$. Hence the largest eigenvalue is

$$\lambda_{\max}(t) = \begin{cases} 0, & t < 0, \\ 2, & t \geq 0. \end{cases}$$

This eigenvalue is not even continuous at $t = 0$, and therefore is not Lipschitz. Explicitly, for any proposed $C \geq 0$, choose $t = -1/(C + 1)$. Then

$$|\lambda_{\max}(t) - \lambda_{\max}(0)| = 2, \quad C|t| = \frac{C}{C + 1} < 1.$$

Thus no finite Lipschitz constant works, irrespective of the bounded insertion degree. The claimed differentiability conclusion also cannot hold at the insertion. Replacing unweighted insertion by a specified smooth weighting would be a different hypothesis absent from the statement.

Formal verification. The Lean project constructs the downward-closed simplicial complexes and proves filtration monotonicity and the edge-insertion criterion. Its boundary matrices use actual edge indices of size zero or one, rather than a fictitious existing edge. It forms the Hodge matrix by multiplying the boundary by its transpose, evaluates its entries, proves the eigenvalue constraints and largest-eigenvalue characterization, and negates every Lipschitz bound for that actual largest eigenvalue. The proof assumes no spectral or boundary certificate; final axiom audits use only standard Lean axioms.